



4 – Proofs and Counterreg.

Week 5 – 9 Sep 2025

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Cheat sheet

r sufficient condition s

$$r \rightarrow s$$

r necessary condition s

$$\neg r \rightarrow \neg s$$

r necc & suff condition s

$$r \leftrightarrow s$$

r unless s

$$\neg s \rightarrow r$$

Cheat sheet

$$\begin{aligned} & \neg \forall x \exists y P(x, y) \\ \equiv & \exists x \neg \exists y P(x, y) \\ \equiv & \exists x \forall y \neg P(x, y) \end{aligned}$$

When we use $\neg$, we flip $\forall$ to $\exists$, vice-versa

Same for $\forall$ and $\wedge$

Cheat sheet

$\forall x \exists y P(x, y) \rightarrow$

- ▷ Given x
- ▷ Find $y = f(x)$ that satisfy $P(x, y)$
eg. $y = x^2$

$\exists x \forall y P(x, y) \rightarrow$

- ▷ Suppose $x = x$, ^{eg.} $x = 2$
- ▷ Give y is true for x ,

Ex. 3.1.8. Let $B(x)$ be “ $-10 < x < 10$ ”. Find the truth set of $B(x)$ for each of the following domains:

- (a) $\mathbb{Z}$ (b) $\mathbb{Z}^+ = \{n \in \mathbb{Z} \mid n > 0\}$ (c) The set of even integers.

SOLUTION . (a) Truth set is $\{x \in \mathbb{Z} \mid B(x)\} = \{x \in \mathbb{Z} \mid -10 < x < 10\} = \{-9, -8, -7, -6, \dots, 6, 7, 8, 9\}$.

(b) Truth set is $\{x \in \mathbb{Z}^+ \mid B(x)\} = \{x \in \mathbb{Z}^+ \mid -10 < x < 10\} = \{1, 2, 3, 4, 5, 6, 7, 8, 9\}$.

(c) Truth set is $\{x \in \text{set of even integers} \mid B(x)\} = \{x \in \text{set of even integers} \mid -10 < x < 10\} = \{-8, -6, -4, -2, 0, 2, 4, 6, 8\}$. Remember that even integers include negative even numbers like $-2, -4, -6, \dots$ as well as 0. □

Ex. 3.1.12. Find counterexamples to show that the statement is false.

$$\forall \text{ real numbers } x \text{ and } y, \sqrt{x+y} = \sqrt{x} + \sqrt{y}.$$

SOLUTION . Counterexample 1: Let $x = 1$ and $y = 1$, and note that

$$\sqrt{1+1} = \sqrt{2} \neq 2 = 1+1 = \sqrt{1} + \sqrt{1}.$$

(Any real numbers x and y with $xy \neq 0$ will produce a counterexample.)

Counterexample 2: Let $a = 1, b = -1$. Then $\sqrt{a+b} = \sqrt{0} = 0$, is a real number. However, $\sqrt{a} + \sqrt{b} = \sqrt{1} + \sqrt{-1}$ is not a real number. □

Ex. 3.1.30. Rewrite each statement without using quantifiers or variables. Indicate which are true and which are false, and justify your answers as best as you can.

Let the domain of x be the set $\mathbb{Z}$ of integers, and let $\text{Odd}(x)$ be “ x is odd”, $\text{Prime}(x)$ be “ x is prime”, and $\text{Square}(x)$ be “ x is a perfect square”. (An integer is a **perfect square** if and only if it equals the square of some integer. For example, 25 is a perfect square because $25 = 5^2$.)

- a. $\exists x$ such that $\text{Prime}(x) \wedge \neg \text{Odd}(x)$.
- b. $\forall x, \text{Prime}(x) \rightarrow \neg \text{Square}(x)$.
- c. $\exists x$ such that $\text{Odd}(x) \wedge \text{Square}(x)$.

SOLUTION . a. This statement translates as “There is a prime number that is not odd.”

This is **true**. The number 2 is prime and it is not odd.

- b. This statement translates as “If an integer is a prime number, then it is not a perfect square.” This is **true**.

Let n be a prime number. Then $n > 1$ and n is not a product of two smaller positive integers. So it cannot be a perfect square.

- c. This statement translates as “There is a number that is both an odd number and a perfect square.” This is **true**. For example, the number 9 is odd and it is also a perfect square (because $9 = 3^2$).



Ex. 3.2.15. Let $D = \{-48, -14, -8, 0, 1, 3, 16, 23, 26, 32, 36\}$. Determine which of the following statements are true and false, and justify your answer.

(b) $\forall x \in D$, if x is less than 0 then x is even.

(d) $\forall x \in D$, if the ones digit of x is 2, then the tens digit of x is 4 or 5.

Ex. 3.2.17, 27. Write a negation, the contrapositive, converse, and inverse of the following statement.

$\forall$ integers d , if $6/d$ is an integer then $d = 3$.

SOLUTION . Negation: $\exists$ an integer d such that $6/d$ is an integer and $d \neq 3$.

Contrapositive: $\forall$ integers d , if $d \neq 3$ then $6/d$ is not an integer.

Converse: $\forall$ integers d , if $d = 3$ then $6/d$ is an integer.

Inverse: $\forall$ integers d , if $6/d$ is not an integer then $d \neq 3$.



Ex. 3.2.47. Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite the following statement without using the words *necessary* and *sufficient*.

The absence of error messages during translation of a computer program is only a necessary and not a sufficient condition for reasonable [program] correctness.

r sufficient condition s

$$r \rightarrow s$$

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$$\neg r \rightarrow \neg s$$

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r necessary condition s

$$\neg r \rightarrow \neg s \equiv s \rightarrow r$$

$\forall P$, if P is correct, then P have no errors

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r

r sufficient condition s

s

$r \rightarrow s$

$$\neg \forall P (r \rightarrow s) \equiv \exists P \neg (r \rightarrow s) \equiv \exists P \neg (\neg r \vee s) \equiv \exists P, r \wedge \neg s$$

$\exists P$ having no errors AND P is incorrect

Ex. 3.2.47. Use the facts that the negation of a $\forall$ statement is a $\exists$ statement and that the negation of an if-then statement is an *and* statement to rewrite the following statement without using the words *necessary* and *sufficient*.

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SOLUTION . Formal Version: $\forall$ computer programs P , if P is correct then P translates without error messages. However, $\exists$ a computer program P such that P translates without error messages and P is not correct.

Ex. 3.3.12. Let $D = E = \{-2, -1, 0, 1, 2\}$. Write negations for each of the following statements and determine which is true, the given statement or its negation.

- a. $\forall x \in D, \exists y \in E$ such that $x + y = 1$.
b. $\exists x \in D$ such that $\forall y \in E, x + y = -y$.

SOLUTION . a. $\exists x \in D$ such that $\forall y \in E, x + y \neq 1$.

The negation is true. Take $x = -2$. Then for any y in E , $x + y < 1$.

- b. $\forall x \in D, \exists y \in E$ such that $x + y \neq -y$.

The negation is true. No matter what number x is given, we may choose a y so that $x + y \neq -y$.

Given x	Choose y	$x + y$	$-y$
-2	0	-2	0
-1	0	-1	0
0	1	1	-1
1	0	1	0
2	0	2	0



Ex. 3.3.35, 36. (a) Rewrite the statement formally using quantifiers and variables, and (b) write a negation for the statement.

35. Everybody trusts somebody.

36. Somebody trusts everybody.

SOLUTION . 35. (a): $\forall$ people x , $\exists$ a person y such that x trusts y .

(b): Negation: $\exists$ a person x such that $\forall$ people y , x does not trust y .

Or: Somebody trusts nobody.

36. (a): $\exists$ a person x such that $\forall$ people y , x trusts y .

(b): Negation: $\forall$ people x , $\exists$ a person y such that x does not trust y .

Or: Nobody trusts everybody.



Ex. 3.3.58. Let $P(x)$ and $Q(x)$ be predicates and suppose D is the domain of x . For the statement forms in each pair, determine whether (a) they have the same truth value for every choice of $P(x)$, $Q(x)$, and D , or (b) there is a choice of $P(x)$, $Q(x)$, and D for which they have opposite truth values.

$$\exists x \in D, (P(x) \wedge Q(x)), \quad \text{and} \quad (\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x)).$$

SOLUTION . They do not necessarily have the same truth values.

Take $D = \mathbb{R}$, and let $P(x)$ be “ x is positive” and let $Q(x)$ be “ x is negative”.

Then “ $\exists x \in D, (P(x) \wedge Q(x))$ ” can be written “ $\exists$ a real number x such that x is both positive and negative,” which is false.

On the other hand, “ $(\exists x \in D, P(x)) \wedge (\exists x \in D, Q(x))$ ” can be written “ $\exists$ a real number that is positive and $\exists$ a real number that is negative”, which is true.

Other possible choices:

$D = \mathbb{R}$ and $P(x)$ be “ $x = 1$ ” and $Q(x)$ be “ $x \neq 1$ ”.

$D = \mathbb{Z}$ and $P(x)$ be “ x is odd” and $Q(x)$ be “ x is even”.

Or you can have your own answer!

